

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE				
Supplementary Winter Examination – 2024				
Course: B.Tech		Branch :Chemical Engg		Semester :VI
Subject Code & Name: BTCHC602 Mass Transfer Operation-II				
Max Marks: 60		Date: 20/12/2024		Duration: 3 Hr.
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.				
		(Level /CO)	Marks	
Q. 1	Objective type questions. (Compulsory Question)			12
1	In distillation where q is defined as the moles of liquid flow in the stripping section per mole of feed introduced, for saturated liquid feed		1/1	1
	a. $q > 1$	b. $q < 1$	c. $q = 1$	d. $q = 0$
2	Rayleigh equation applies to _____ distillation		1/1	1
	a. Differential	b. Flash	c. Equilibrium	d. Molecular
3	The relative volatility of a binary mixture at the Azeotropic composition is		1/1	1
	a. 1	b. > 1	c. 0	d. ∞
4	In case of cooling towers, the ratio of the rates of heat and mass transfer is indicated by the _____ number.		1/2	1
	a. Lewis	b. Grashoff	c. Sherwood	d. None of these
5	The wet and 'dry bulb temperature for a vapour-gas mixture are 25°C and 30°C respectively. If the mixture is heated to 45°C at constant pressure, the wet bulb temperature will be _____ °C.		1/2	1
	a. 25	b. > 25	c. < 25	d. - 25
6	Moisture content of a substance when at equilibrium with a given partial pressure of the vapour is called the _____ moisture.		1/3	1
	a. Free	b. Unbound	c. Equilibrium	d. Bound
7	Drying of a solid involves _____ transfer		1/3	1
	a. Only heat	b. Only mass	c. Both heat and mass	d. None of these
8	The critical moisture content in case of drying indicates the _____ rate period.		1/3	1
	a. Beginning of falling	b. Beginning of constant	c. End of falling	d. None of these
9	Separation of two volatile liquids by distillation makes use of their		1/1	1
	a. Selectivity	b. Relative volatility	c. Solubility	d. Density difference
10	Minimum reflux ratio in a distillation column results in		1/1	1
	a. Optimum number of trays	b. Minimum reboiler size	c. Maximum condenser size	d. Minimum number of trays
11	What is crystallizer used for?		1/4	1
	a. Liquid-solid separation	b. Solid separation	c. Liquid – liquid separation	d. Gas – solid separation
12	What is the purpose of seed crystal in crystal growth?		1/4	1

	a. Nucleation	b. Catalyst	c. Solvent	d Solution																										
Q. 2	Solve the following.					12																								
A)	Describe the operation of flash distillation unit and state the equation of operating line.				3/1	6																								
B)	Explain Simple (Differential) distillation operation with neat sketch. Write down Rayleigh equation and state its significance.				3/1	6																								
Q.3	Solve the following.					12																								
A)	Describe continuous fractional distillation operation with neat sketch. Explain importance of optimum reflux for operation of column.				3/1	6																								
B)	A mixture of benzene and toluene containing 38 mole % of benzene is to be separated to give a product of 90 mole % benzene at the top, and the bottom product with 4 mole % benzene. The feed enters the column at its boiling point and vapour leaving the column is simply condensed and provide product and reflux. It is proposed to operate the unit with a reflux ratio of 3.0. Equilibrium data for benzene as component as follow				4/1	6																								
	<table><tr><td>x</td><td>0</td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.4</td><td>0.5</td><td>0.6</td><td>0.7</td><td>0.8</td><td>0.9</td><td>1</td></tr><tr><td>y</td><td>0</td><td>0.22</td><td>0.38</td><td>0.52</td><td>0.63</td><td>0.71</td><td>0.79</td><td>0.85</td><td>0.91</td><td>0.96</td><td>1</td></tr></table>				x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1	y	0	0.22	0.38	0.52	0.63	0.71	0.79	0.85	0.91	0.96	1		
x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1																			
y	0	0.22	0.38	0.52	0.63	0.71	0.79	0.85	0.91	0.96	1																			
	Find out theoretical number of plates required to carry out this operation using McCabe Thiele's method.																													
Q. 4	Solve Any Two of the following.					12																								
A)	Describe any two types of cooling tower used in chemical industry with neat sketch				2/2	6																								
B)	How psychometric chart does help in humidification-dehumidification operation.				2/2	6																								
C)	Air–water vapour mixture has a DBT of 55°C with humidity of 0.048 kmol water vapour/kmol dry air and 1 standard atmospheric pressure. Find absolute humidity, humid heat and total enthalpy.				3/2	6																								
Q.5	Solve Any Two of the following.					12																								
A)	Describe any two types of dryers used in chemical industry for granular solids				2/3	6																								
B)	Explain how moisture content and drying rate changes with respect to drying time for a moist slab with the help of drying rate-moisture content graph.				3/3	6																								
C)	Wet solid having 1000 kg dry weight is dried under constant drying conditions with an air velocity of 0.75 m/s, so that the surface area of drying is 55 m ² . The critical moisture content of the material is 0.125 kg water/kg dry solids. If the initial rate of drying is 0.3 x 10 ⁻³ kg/m ² s. How long will it take to dry the material from 0.15 to 0.025 kg water/kg dry solid?				4/3	6																								
Q. 6	Solve Any Two of the following.					12																								
A)	Explain in detail how crystallization operation is carried out using Forced circulation crystallizer.				3/4	6																								
B)	Which different types of membrane separation methods are used in chemical industry? How do they differ?				2/4	6																								
C)	What do you understand from crystal formation theories? What are the factors on which crystallization depends?				3/4	6																								
	*** End ***																													